



WORLD REMIT PAYMENT PROCESSOR RE- LEASE NOTES

DECEMBER 09, 2020

RELEASE DESCRIPTION AND CHANGE HISTORY

Product Name	Version (Major. Minor)	Release (Initial, Minor, Fix & Hot fix), Date	Date	Author	Reviewed By
World Remit - Payment Processor	1.0	Initial	December 09, 2020	Kennedy Gatimu	John Muli

RELEASE PREREQUISITES

#	Prerequisites
1	Database
2	Kafka
3	Connectivity to World Remit
4	Connectivity to GAG/ TPG
5	Self-signed certificate for GAG authentication

RELEASE OVERVIEW

This release notes contains the working of the payment processor:

1. Database Access
2. Database Tables Description
3. GAG/ TPG operations
4. Monitoring
5. Production of self-signed certificate

1. Database Access

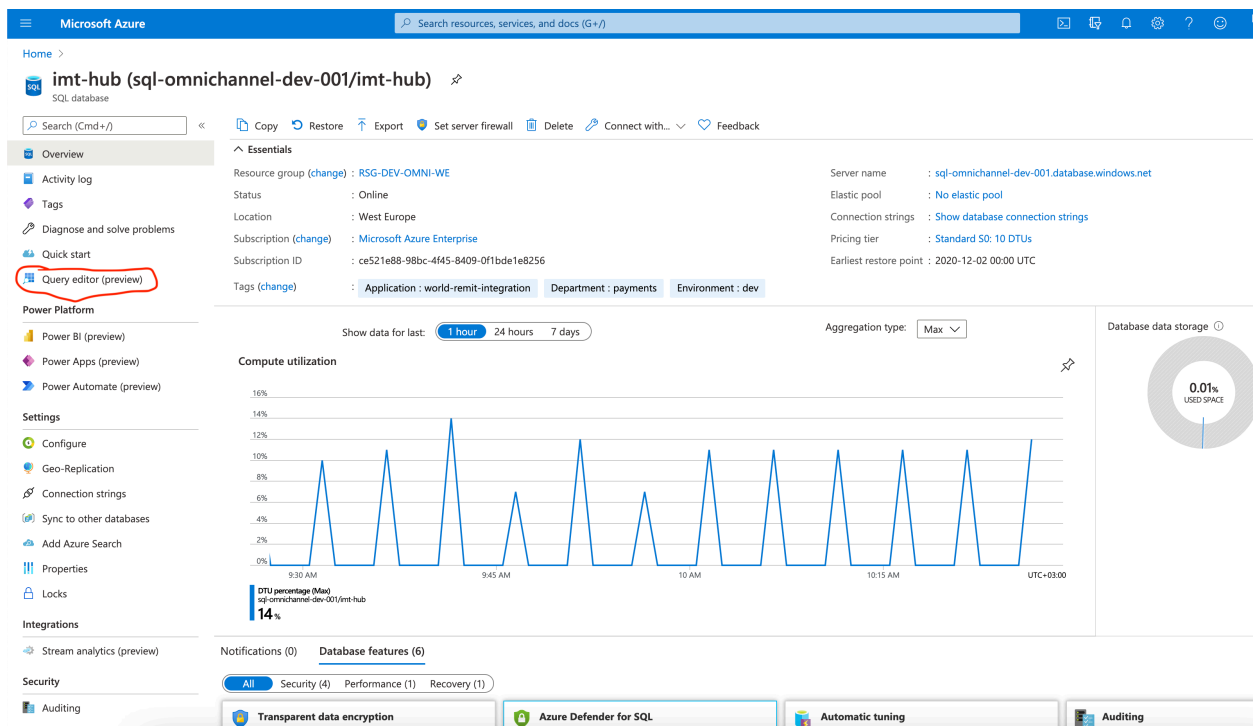
1.1 A new DB needs to be created in cloud.

DB Name: **imt-hub**

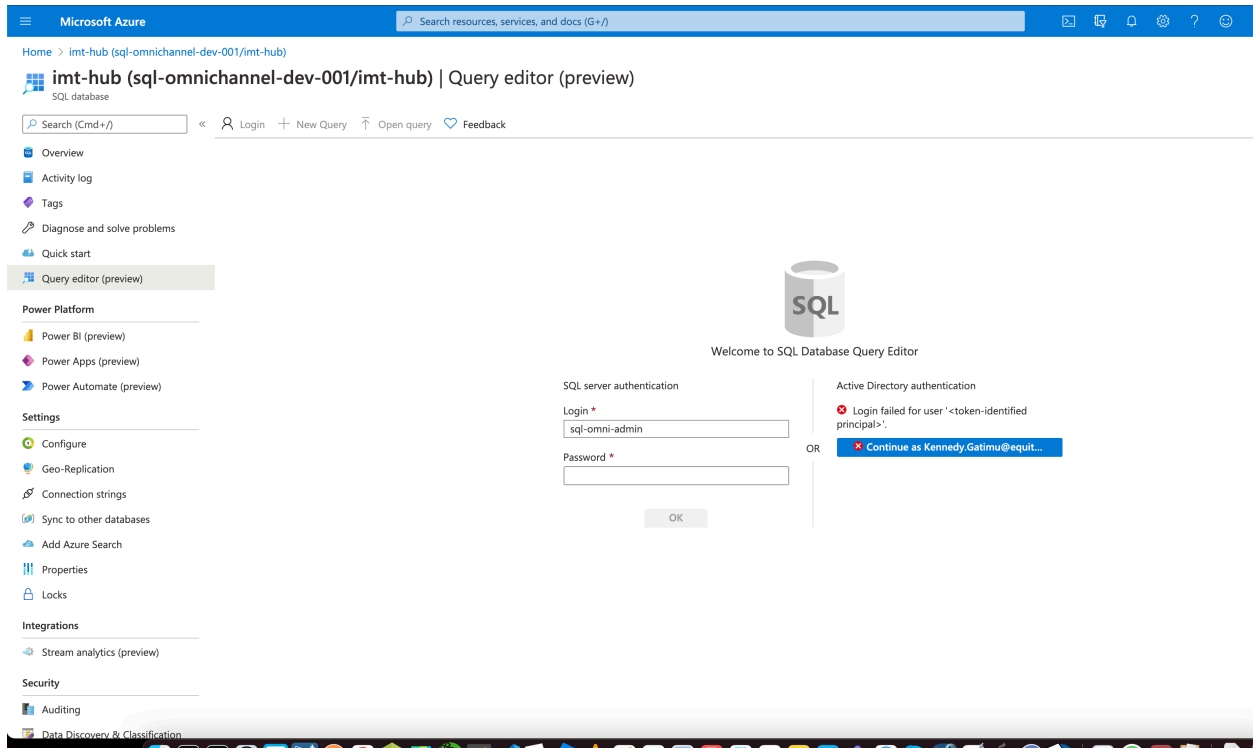
1.2 After DB has been created:

1.2.1 Go to the link: <https://portal.azure.com/#@equitybank.co.ke/resource/subscriptions/ce521e88-98bc-4f45-8409-0f1bde1e8256/resourceGroups/RSG-DEV-OMNI-WE/providers/Microsoft.Sql/servers/sql-omnichannel-dev-001/databases/imt-hub/queryEditor>

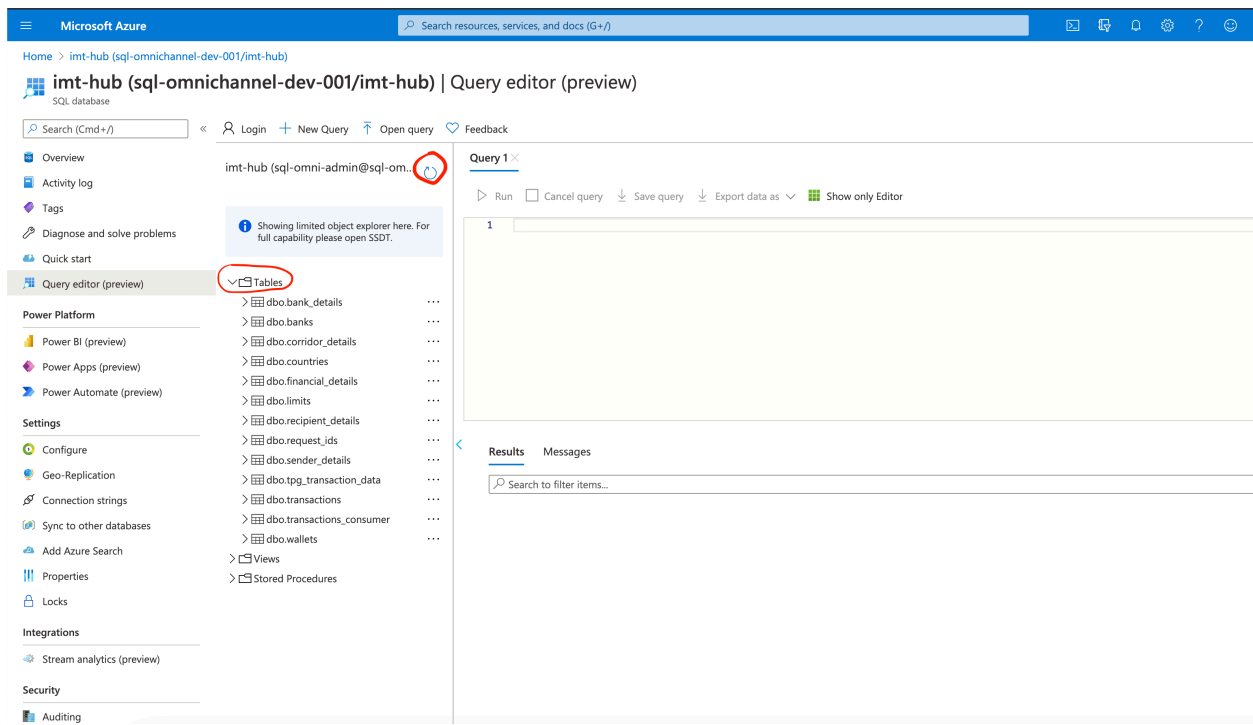
ps. This is UAT link, we should have a production link for this case



1.2.2 Click on Query editor (preview)



1.2.3 Enter the provided credentials



1.2.4 Click refresh and expand tables as shown above

2. Database Tables Description

Below are the tables to be created

2.1 bank_details

```
CREATE TABLE `bank_details` (  
  `id` bigint NOT NULL,  
  `account_number` varchar(255) DEFAULT NULL,  
  `bic` varchar(255) DEFAULT NULL,  
  `bank_code` varchar(255) DEFAULT NULL,  
  `bank_name` varchar(255) DEFAULT NULL,  
  PRIMARY KEY (`id`)  
)
```

2.2 banks

```
CREATE TABLE `banks` (  
  `id` bigint NOT NULL,  
  `code` varchar(50) NOT NULL,  
  `name` varchar(50) NOT NULL,  
  `country_id` bigint NOT NULL,  
  `limit_id` bigint DEFAULT NULL,  
  PRIMARY KEY (`id`),  
  KEY `FKlmiqijve14o6vd15ttic40qnc` (`country_id`),  
  KEY `FKlvad6w17nhw1mp7gak886ojiu` (`limit_id`),  
  CONSTRAINT `FKlmiqijve14o6vd15ttic40qnc` FOREIGN KEY (`country_id`) REFER-  
ENCES `countries` (`id`),  
  CONSTRAINT `FKlvad6w17nhw1mp7gak886ojiu` FOREIGN KEY (`limit_id`) REFER-  
ENCES `limits` (`id`)  
)
```

2.3 corridor_details

```
CREATE TABLE `corridor_details` (  
  `id` bigint NOT NULL,  
  `api_code` varchar(255) DEFAULT NULL,  
  `recipient_country_code_iso2` varchar(255) DEFAULT NULL,
```

```

`recipient_country_dial_code` varchar(255) DEFAULT NULL,
`recipient_currency_code` varchar(255) DEFAULT NULL,
`settlement_currency_code` varchar(255) DEFAULT NULL,
PRIMARY KEY (`id`)
)

```

2.4 countries

```

CREATE TABLE `countries` (
  `id` bigint NOT NULL,
  `name` varchar(50) NOT NULL,
  PRIMARY KEY (`id`),
  UNIQUE KEY `UK_1pyiwrqimi3hnl3vtgsypj5r` (`name`)
)

```

2.5 financial_details

```

CREATE TABLE `financial_details` (
  `id` bigint NOT NULL,
  `settlement_amount` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`id`)
)

```

2.6 limits

```

CREATE TABLE `limits` (
  `id` bigint NOT NULL,
  `currency` varchar(255) DEFAULT NULL,
  `max` decimal(19,2) DEFAULT NULL,
  `min` decimal(19,2) DEFAULT NULL,
  `bank_id` bigint DEFAULT NULL,
  `wallet_id` bigint DEFAULT NULL,
  PRIMARY KEY (`id`),
  KEY `FKgt9jrsh7a6ee3i51o8sqyv74e` (`bank_id`),
  KEY `FKhjbmcsp6x0ntpuyoj8ctmg89q` (`wallet_id`),
  CONSTRAINT `FKgt9jrsh7a6ee3i51o8sqyv74e` FOREIGN KEY (`bank_id`) REFERENCES
`banks` (`id`),

```



```
CONSTRAINT `FKhjbmcsp6x0ntpuyoj8ctmg89q` FOREIGN KEY (`wallet_id`) REFER-  
ENCES `wallets` (`id`)  
)
```

2.7 recipient_details

```
CREATE TABLE `recipient_details` (  
  `id` bigint NOT NULL,  
  `first_name` varchar(255) DEFAULT NULL,  
  `last_name` varchar(255) DEFAULT NULL,  
  `middle_name` varchar(255) DEFAULT NULL,  
  `mobile_account_number` varchar(255) DEFAULT NULL,  
  PRIMARY KEY (`id`)  
)
```

2.8 request_ids

```
CREATE TABLE `request_ids` (  
  `id` bigint NOT NULL,  
  `date_created` varchar(255) DEFAULT NULL,  
  `date_updated` varchar(255) DEFAULT NULL,  
  `request_id` varchar(255) NOT NULL,  
  PRIMARY KEY (`id`),  
  UNIQUE KEY `request_id` (`request_id`)  
)
```

2.9 sender_details

```
CREATE TABLE `sender_details` (  
  `id` bigint NOT NULL,  
  `first_name` varchar(255) DEFAULT NULL,  
  `last_name` varchar(255) DEFAULT NULL,  
  `middle_name` varchar(255) DEFAULT NULL,  
  PRIMARY KEY (`id`)  
)
```

2.10 tpg_transaction_data

```
CREATE TABLE `tpg_transaction_data` (  
  `id` bigint NOT NULL,  
  `log_date` varchar(255) DEFAULT NULL,  
  `product_code` varchar(255) DEFAULT NULL,  
  `reference` varchar(255) DEFAULT NULL,  
  `tpg_code` varchar(255) DEFAULT NULL,  
  `tpg_message` varchar(255) DEFAULT NULL,  
  `tpg_request` text,  
  `tpg_response` text,  
  `tpg_status` varchar(255) DEFAULT NULL,  
  `tpg_tran_ref` varchar(255) DEFAULT NULL,  
  `wr_response_code` varchar(255) DEFAULT NULL,  
  `wr_response_msg` varchar(255) DEFAULT NULL,  
  PRIMARY KEY (`id`),  
  UNIQUE KEY `reference` (`reference`),  
  UNIQUE KEY `UK_4evw89nl78s7jqku75n289678` (`reference`)  
)
```

2.11 transactions

```
CREATE TABLE `transactions` (  
  `id` bigint NOT NULL,  
  `date_created` varchar(255) DEFAULT NULL,  
  `date_updated` varchar(255) DEFAULT NULL,  
  `message` varchar(255) DEFAULT NULL,  
  `product_code` varchar(255) DEFAULT NULL,  
  `reference` varchar(255) NOT NULL,  
  `status` varchar(255) DEFAULT NULL,  
  `world_remit_id` varchar(255) NOT NULL,  
  `wrid` varchar(255) DEFAULT NULL,  
  `bank_detail_id` bigint DEFAULT NULL,  
  `corridor_detail_id` bigint DEFAULT NULL,  
  `financial_detail_id` bigint DEFAULT NULL,
```

```

`recipient_detail_id` bigint DEFAULT NULL,
`sender_detail_id` bigint DEFAULT NULL,
PRIMARY KEY (`id`),
UNIQUE KEY `reference` (`reference`),
UNIQUE KEY `world_remit_id` (`world_remit_id`),
KEY `FKj0lyw66uiii5hhc4mw05v510` (`bank_detail_id`),
KEY `FKjbf9vikusth0igse56ip0cn66` (`corridor_detail_id`),
KEY `FK73rwte0ovwvjey0fxx2x6axtv` (`financial_detail_id`),
KEY `FKkjio9wkhewhg05e4gjkhisuh9` (`recipient_detail_id`),
KEY `FKmiccificqwu3x06ulphc2vla1` (`sender_detail_id`),
CONSTRAINT `FK73rwte0ovwvjey0fxx2x6axtv` FOREIGN KEY (`financial_detail_id`) REFERENCES `financial_details` (`id`),
CONSTRAINT `FKj0lyw66uiii5hhc4mw05v510` FOREIGN KEY (`bank_detail_id`) REFERENCES `bank_details` (`id`),
CONSTRAINT `FKjbf9vikusth0igse56ip0cn66` FOREIGN KEY (`corridor_detail_id`) REFERENCES `corridor_details` (`id`),
CONSTRAINT `FKkjio9wkhewhg05e4gjkhisuh9` FOREIGN KEY (`recipient_detail_id`) REFERENCES `recipient_details` (`id`),
CONSTRAINT `FKmiccificqwu3x06ulphc2vla1` FOREIGN KEY (`sender_detail_id`) REFERENCES `sender_details` (`id`)
)

```

2.12 transactions_consumer

```

CREATE TABLE `transactions_consumer` (
  `id` bigint NOT NULL,
  `body` text,
  `message` varchar(255) DEFAULT NULL,
  `reference` varchar(255) DEFAULT NULL,
  `status` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`id`)
)

```

2.13 wallets

```
CREATE TABLE `wallets` (  
  `id` bigint NOT NULL,  
  `name` varchar(50) NOT NULL,  
  `country_id` bigint NOT NULL,  
  `limit_id` bigint DEFAULT NULL,  
  PRIMARY KEY (`id`),  
  UNIQUE KEY `UK_21qj4jydkkp1tnmyd3ektcjl4` (`name`),  
  KEY `FK1py3hlay4qqpl1567pusso66o` (`country_id`),  
  KEY `FKrbvoqt67vnp0q8rd8c9r2p3bn` (`limit_id`),  
  CONSTRAINT `FK1py3hlay4qqpl1567pusso66o` FOREIGN KEY (`country_id`) REFERENCES `countries` (`id`),  
  CONSTRAINT `FKrbvoqt67vnp0q8rd8c9r2p3bn` FOREIGN KEY (`limit_id`) REFERENCES `limits` (`id`)  
)
```

3. GAG/ TPG operations

GAG/ TPG operations will be as follows:

1. Bank Transfers

- Send Money to Equity Bank KE Account
- Send Money to Equity RTGS KE Account
- Send Money to Equity Pesalink KE Account
- Send Money to Equity Bank TZ Account
- Send Money to Equity RTGS TZ Account
- Send Money to Equity Bank UG Account
- Send Money to Equity Bank RW Account

2. Mobile Money

- Send Money to Mpesa KE
- Send Money to Airtel KE
- Send Money to Airtel TZ
- Send Money to Vodacom TZ
- Send Money to MTN UG
- Send Money to MTN RW
- Send Money to Tigo RW

3.1 Bank Transfers

3.1.1 Send Money to Equity Bank KE Account

Request:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
```

```
</S:Header>
<S:Body>
  <ns2:tranApplyRequest>
    <accountNumber>0810197039302</accountNumber>
    <paymentReference1>27a043422e41</paymentReference1>
    <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
    <paymentReference3>27a043422e41</paymentReference3>
    <paymentType>WorldRemit</paymentType>
    <remarks>Jomo Joshua Kiplimo</remarks>
    <agentID>2448</agentID>
    <check>Vt5Js7B1mAHLSQQm8IfOP4dpCYbhJqcD//czT8JFULxuQ7w4uD2kSg-
RuVKd8fswcWGfCbt6etahWgvS0rnlcGFQ/X76OsdHfLIUFvT/9iF2INQ+BFshezfdqJH6p-
HLzDYyz08WsRStdR/S3iYHwuNBT6ZaFfrhPk3/vvGh9zVTTs+N6XKSKUAvAMR+DY3D-
M7VQs3jUYQEtAw6bZkA6V+j/yk+MabTu7iz5bMTIRHepCLTch5YXfQI0mnZXpkNN/
Lf9qDCTOiKRllKslm2zPx1GRYp4OwZHiwaGA88UfvNLElFXlNR5nY4FG86bmrSSw94+0K
hJvRTTK9eEXvbH4K7A==</check>
    <fees>0.0</fees>
    <instCode>KE</instCode>
    <tranAmount>1000.00</tranAmount>
    <messageID>2781ae043441</messageID>
    <currencyCode>KES</currencyCode>
    <countryCode>KE</countryCode>
    <bicCode>EQBLKENA</bicCode>
    <senderName>Andrii Ol Borysenko</senderName>
    <mobileNumber></mobileNumber>
    <walletName></walletName>
  </ns2:tranApplyRequest>
</S:Body>
</S:Envelope>
```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|060894113847</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>

```

3.1.2 Send Money to Equity RTGS KE Account

Request:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber>12345678000</accountNumber>
      <paymentReference1>qw345j5666116</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
      <paymentReference3>qw345j5666116</paymentReference3>
      <paymentType>WorldRemit</paymentType>
      <remarks>Jomo Joshua Kiplimo</remarks>
      <agentID>2448</agentID>
      <check>RWtbjxgFI56Zrk/k1g3ZWJJNlbSuMK3enK43/6hEJHaKwDq2yn34-
DOAMtvCeszUZJaL+Fi2IAIGqYPvQ1RNbMNnPFVa+AiU0i4TTn2GsIrmPwHN/

```

bqCf3tlp2J8ik9MpenPONpDKguh9b4nV1ZVJwudWUItLCDYrYIA7BOuXeGZIU8Fm-
mu0yWRyY+fn+4G4rPd6Bz3TF2qrDYAWeV3svafBoD99IS4RSx5EY+I08U8a2fsuXBa-
BlklBKzMu4gg/tzqOcd75kkakjvIUt9fKa7Vyf1Yf9Ojy5grmUZTyWaEasQ5wQtOO4BB-
D67lu4NWKeOmr+EQMAXJ0P9KGxFq7r4g==</check>

```
<fees>0.0</fees>
<instCode>KE</instCode>
<tranAmount>1000.00</tranAmount>
<messageID>27812e341151</messageID>
<currencyCode>KES</currencyCode>
<countryCode>KE</countryCode>
<bicCode>BARBKENA</bicCode>
<senderName>Andrii Ol Borysenko</senderName>
<mobileNumber></mobileNumber>
<walletName></walletName>
</ns2:tranApplyRequest>
</S:Body>
</S:Envelope>
```

Response:

```
<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|000002363955</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>
```

3.1.3 Send Money to Equity Pesalink KE Account


```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber>12345678000</accountNumber>
      <paymentReference1>asde23133334323</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
      <paymentReference3>asde23133334323</paymentReference3>
      <paymentType>WorldRemit</paymentType>
      <remarks>Jomo Joshua Kiplimo</remarks>
      <agentID>2448</agentID>
      <check>YtTCsOY9vOX52e8KaxQJKQ3rdp3e2g6+qJzwzYOMsguY6E96QePleS6i/
3xkwP3tJse2qqZ18Al83HNHPEWq68N4FwB+zR4USl2259ofMGSIy28mtk0xrP1PDJr-
WmkM9jgE9pKKkIO325PTvB9dO77fY38rOMUcpKX3veasWmL+Z/HpiltJ5a1z972z7zj2-
Klc65lzpW61OBvXE4secB+s8bD0xuCkcU4+oxWYU0CswkwSWDu7dbbN3lFm4VM+e2t
8bR4D+mYBUm+ylLAGBPnAn/adOlc2/ZoBDtfzPH3CSOdYl6K2lX+DERAQqqfC1lyu/
8JrAWkLGwlrGQ+W06Cg==</check>
      <fees>0.0</fees>
      <instCode>KE</instCode>
      <tranAmount>15000.00</tranAmount>
      <messageID>34456q09755</messageID>
      <currencyCode>KES</currencyCode>
      <countryCode>KE</countryCode>
      <bicCode>CITIKENA</bicCode>
      <senderName>Andrii Ol Borysenko</senderName>
      <mobileNumber></mobileNumber>
      <walletName></walletName>
    </ns2:tranApplyRequest>
  </S:Body>
</S:Envelope>

```

```
    </ns2:tranApplyRequest>
  </S:Body>
</S:Envelope>
```

Response:

```
<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|000002363956</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>
```

3.1.4 Send Money to Equity Bank TZ Account

Request:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber>3008111195295</accountNumber>
      <paymentReference1>1we51341115</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
      <paymentReference3>1we51341115</paymentReference3>
```

```

<paymentType>WorldRemit</paymentType>
<remarks>Jomo Joshua Kiplimo</remarks>
<agentID>2448</agentID>
  <check>FUKyhvNcHyMd/RxUCi10F07AFb7ALnMQ3Mf7iezNjCh3sYhy+1xkdyN-
HJfN+ATx09AnNa57pnZ9pU2Fy9rYb3ON6n2l2s8VT1SoSONsJybe4uKD5xl-
r9bte6Jhqm0zEve8DxG5skHI01LJBxJ+2NIzq6uUz4OXhxpT8aU09W9T0IKG7O8ZTiw+lo
2JL206p7m9aCZSlkKBpUouQ+AbtWCGEgIO8dF2MNeAc8wB78kR8mn1okwmuxiT-
t1gbqDTjDIsL7Jd2tvK00iLncSyHxDwjDhyZTYImK2zriOwATL9J87lgBQhmLIWslxfdtZM+j
auOr2+rnZKWpy3ebxeZOGKw==</check>
  <fees>0.0</fees>
  <instCode>TZ</instCode>
  <tranAmount>10034.56</tranAmount>
  <messageID>1212wex1115</messageID>
  <currencyCode>TZS</currencyCode>
  <countryCode>TZ</countryCode>
  <bicCode>EQBLTZTZXXX</bicCode>
  <senderName>Andrii Ol Borysenko</senderName>
  <mobileNumber></mobileNumber>
  <walletName></walletName>
</ns2:tranApplyRequest>
</S:Body>
</S:Envelope>

```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|074983115700</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>

```

</soap:Envelope>

3.1.5 Send Money to Equity RTGS TZ Account

Request:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber>0187150072001</accountNumber>
      <paymentReference1>90987ww1311</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
      <paymentReference3>90987ww1311</paymentReference3>
      <paymentType>WorldRemit</paymentType>
      <remarks>Jomo Joshua Kiplimo</remarks>
      <agentID>2448</agentID>
      <check>bAKZX24z9Cs587Q7VetkX/gSdxQymmCofCrt2hPjHS1ZZLnuu/54Tdkjy-
        IEI3hWYIN4SqjLLld8xtMFUJRd290lsHlnSbqOzQp2Zc/HqYjgXH+qtCotL3VlzHKt-
        Pwx3LDp9RHOMo8coUhntw16dR8uBwwAmssmBtEig8JXyzx/d5taqB6GpgU9/YyaQU/
        VX4BSid/vLSSqlbdtSlbJzEeGtS3Vos4HdSlxeqApjPTx/7Mhm8jHKLvaCTf4PCd2YKgZkP-
        kZs+MDSkEwctVRIFJAYJBnTbtM0167dCKDfj3AvjMlx4TjbouEcONIXcpv2TwUNbgBijg-
        IDmTpqxEhw6yg==</check>
      <fees>0.0</fees>
      <instCode>TZ</instCode>
      <tranAmount>10004.56</tranAmount>
      <messageID>990000228765</messageID>
      <currencyCode>TZS</currencyCode>
```

```

    <countryCode>TZ</countryCode>
    <bicCode>DTKETZTZ</bicCode>
    <senderName>Andrii Ol Borysenko</senderName>
    <mobileNumber></mobileNumber>
    <walletName></walletName>
  </ns2:tranApplyRequest>
</S:Body>
</S:Envelope>

```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|0749235710</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>

```

3.2.6 Send Money to Equity Bank UG Account

Request:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>

```

```

<ns2:tranApplyRequest>
  <accountNumber>1000101365473</accountNumber>
  <paymentReference1>9098311ug</paymentReference1>
  <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
  <paymentReference3>9098311ug</paymentReference3>
  <paymentType>WorldRemit</paymentType>
  <remarks>Jomo Joshua Kiplimo</remarks>
  <agentID>2448</agentID>
  <check>hM6Dewhsz5vAeB9aWiLZW7wM32fOf05Eg2Fpc9h+uw6L2YmDYDs-
J0EbEc3gW9g3AHkvXUVVlgqdnbMVfVqsSSbSVgbTU02Tk73WCQoKNUOeDWK0+Q/
eAHmRDQIWhNScHHdB5oqUTlz2mtJEIposlki5jcltVz3NqPBRMFrVz6XJb4g/yey+vqXa-
cLYmviX13qInDzy9IUe0KZpBtVH3fs684eIO/dPQdRm2M6E8yDry0PQGZo7KDKywbK-
TMZx6IZ0OxdQ+APTLU/O681jMXoR7Uw9eU3oIJoEsWgH88FXK/
GnTajTm+jbL8UCJWs1saBvAzhxLSRCzwr4PPjlnN0kA==</check>
  <fees>0.0</fees>
  <instCode>UG</instCode>
  <tranAmount>10004.56</tranAmount>
  <messageID>9900005ug</messageID>
  <currencyCode>UGX</currencyCode>
  <countryCode>UG</countryCode>
  <bicCode>EQBLUGKA</bicCode>
  <senderName>Andrii Ol Borysenko</senderName>
  <mobileNumber></mobileNumber>
  <walletName></walletName>
</ns2:tranApplyRequest>
</S:Body>
</S:Envelope>

```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>

```

```

<ns2:tranApplyRequestResponse
  xmlns:ns2="http://ws.equitybank/">
  <return>OK|661509120504</return>
</ns2:tranApplyRequestResponse>
</soap:Body>
</soap:Envelope>

```

3.1.7 Send Money to Equity Bank RW Account

Request:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber>4004100433300</accountNumber>
      <paymentReference1>9098311rw</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
      <paymentReference3>9098311rw</paymentReference3>
      <paymentType>WorldRemit</paymentType>
      <remarks>Jomo Joshua Kiplimo</remarks>
      <agentID>2448</agentID>
      <check>VbZRslAerdsNrlj4OJmLA782Vhqk+IrpzxcSeU+Hoy4ZxKaHjmdjtyP+1b-
nWM/mNAyBXHWCMEakyoMfWdtE3p0OACP/TxbxTW1WJeqakVjfwPkhOCV+xsQhs-
Mp5D5CLtILUjdWkw/siJAH01l66a1EINpOWXEktUjulz6OSlchc9f5zd1eQd5+CmnmU5t-
fVojMaFOQYHL5D6Z6FWSXD9hivSgj07r34EzV0p8T51A7yMg2dEsJ/mwBJD/dwB1G-
PSeExAim/I3O7H2kcrYQz9FwoEEfd7wJ9TobeN2IYG0z7PJUX9D3Mw8ssQX-
is9q8XSE7FPgFGP3tSG9ZY4ZLVMvA==</check>
      <fees>0.0</fees>
    </ns2:tranApplyRequest>
  </S:Body>
</S:Envelope>

```

```

    <instCode>RW</instCode>
    <tranAmount>10004.56</tranAmount>
    <messageID>9900005rw</messageID>
    <currencyCode>RWF</currencyCode>
    <countryCode>RW</countryCode>
    <bicCode>EQBLRWRWXXX</bicCode>
    <senderName>Andrii Ol Borysenko</senderName>
    <mobileNumber></mobileNumber>
    <walletName></walletName>
  </ns2:tranApplyRequest>
</S:Body>
</S:Envelope>

```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|036991121050</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>

```

3.2 Mobile Money

3.2.1 Send Money to Mpesa KE

Request:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
```



```

<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber></accountNumber>
      <paymentReference1>23459qw33</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
      <paymentReference3>23459qw33</paymentReference3>
      <paymentType>WorldRemit</paymentType>
      <remarks>Jomo Joshua Kiplimo</remarks>
      <agentID>2448</agentID>
      <check>Ve/ff4c1Wxq6KliDebA7v0WJ0iUiPwYAan8kpFhyfy3KhaKkJgb1Ag6Yxy-
V9gzsD8Lu/wjphYjuKhHsAc+VAu5eNe/IDEJ4TAy8/QydbwzASMvN+N6izmX14qCn7st-
Y44JK4BAYMdwFk0Ko23xaH4n80KGo7JPqRzFCdPRfKdboGHcQTHjVKCAR87jA/twakT-
SOCaL3cPUGuW6TxMKd/uxeAjB/gLp9C9vbhOASld7I7iSX3ucs+YLiFEopk1/PwJlkjQYOl-
gL+rbrzdAP8qhbcETWPYzUqF0SCKb5uE6I21YVVkecl+elizeeGnMSmqG1Qtb1W0SsC-
t6woWNAhLDw==</check>
      <fees>0.0</fees>
      <instCode>KE</instCode>
      <tranAmount>104.00</tranAmount>
      <messageID>23450a00221</messageID>
      <currencyCode>KES</currencyCode>
      <countryCode>KE</countryCode>
      <bicCode></bicCode>
      <senderName>Andrii Ol Borysenko</senderName>
      <mobileNumber>254705912645</mobileNumber>
      <walletName>MPESA</walletName>
    </ns2:tranApplyRequest>
  </S:Body>
</S:Envelope>

```

```
</S:Body>
</S:Envelope>
```

Response:

```
<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|255084121426</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>
```

3.2.2 Send Money to Airtel KE

Request:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string" xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"></S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber></accountNumber>
      <paymentReference1>1123a1111</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
      <paymentReference3>1123a1111</paymentReference3>
      <paymentType>WorldRemit</paymentType>
      <remarks>Jomo Joshua Kiplimo</remarks>
      <agentID>2448</agentID>
```

```

        <check>RN9u6czY0+gFv1t6ZX24Up9qKMjppRUTbNizGKEw2+2GzUt-
LVQN2S0k38Z/WKmzoocY5fDtPRcbGnH6sR0viT4h5dK/PuFV/HcsOCTgdw9OCTI6Kb-
RZTjtLiZbEZvgpcl4Wde7SBCQU+BIMu1bdDRJ1VWkwS0JyhfsrmgBBzXQimRrdG8guf-
RLN+2O/
l1ojiP9vx7er4RSOL8KzSx9E59wZzPojCzhlHwYNDUQeh7eL9MmJUJcVf35q+E6/Ddo-
DMH8J5tqbeUgsiYbOUg71HvLDIEecQuKrhT9lavP6AQJTKbZBY+VO9DLBezMBV5x4qy-
TkOI/MWQNERgEMGt+k89A==</check>

        <fees>0.0</fees>
        <instCode>KE</instCode>
        <tranAmount>104.56</tranAmount>
        <messageID>211113q01001</messageID>
        <currencyCode>KES</currencyCode>
        <countryCode>KE</countryCode>
        <bicCode></bicCode>
        <senderName>Andrii Ol Borysenko</senderName>
        <mobileNumber>254733000000</mobileNumber>
        <walletName>AIRTELMONEY-KE</walletName>
    </ns2:tranApplyRequest>
</S:Body>
</S:Envelope>

```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|2550841223426</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>

```

</soap:Envelope>

3.2.3 Send Money to Airtel TZ

Request:

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>

<S:Envelope

xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"

xmlns:ns2="http://ws.equitybank/">

<S:Header xsi:type="xs:string"

xmlns:xs="http://www.w3.org/2001/XMLSchema"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">

</S:Header>

<S:Body>

<ns2:tranApplyRequest>

<accountNumber></accountNumber>

<paymentReference1>445ttqv</paymentReference1>

<paymentReference2>Jomo Joshua Kiplimo</paymentReference2>

<paymentReference3>445ttqv</paymentReference3>

<paymentType>WorldRemit</paymentType>

<remarks>Jomo Joshua Kiplimo</remarks>

<agentID>2448</agentID>

<check>XwnNGIQFO2sFLRc2HOfFSr6j6y0+qzPCYIFCTEoUMQqpDnACk-S0TJ4OaTYsJza17EleDTZ1Mu/5fwD83Ntl0ISGt85az0pBTHaCSMsQriqN01cUKC-QMv5AZtJVDGycUSD1CmkuuzlLgXCt9bgY1o96CP0G+oyTAPMLkQKawPofy/4lf5SPocxHi1NNb61CxjRHllgMxzngYlnQGbbqVoqjjDtBliBpxspRk/sjtlyME3ODmZ-IZQg8Nkvqo7r6ovWgGnAYNndn4VXNOwlhXkU81uDxGYbbHMZiLB2naz7D-p1Uyj8dBN+to386Lito1LEbMUzF8DgDh6ZDpBWDKIRpcg==</check>

<fees>0.0</fees>

<instCode>TZ</instCode>

<tranAmount>1042.24</tranAmount>

<messageID>12309944827600</messageID>

<currencyCode>TZS</currencyCode>

<countryCode>TZ</countryCode>

```

    <bicCode></bicCode>
    <senderName>Andrii Ol Borysenko</senderName>
    <mobileNumber>255784000000</mobileNumber>
    <walletName>AIRTELMONEY-TZ</walletName>
  </ns2:tranApplyRequest>
</S:Body>
</S:Envelope>

```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|793142121915</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>

```

3.2.4 Send Money to Vodacom TZ

Request:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>

```

```

<ns2:tranApplyRequest>
  <accountNumber></accountNumber>
  <paymentReference1>11203q13q4501</paymentReference1>
  <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
  <paymentReference3>11203q13q4501</paymentReference3>
  <paymentType>WorldRemit</paymentType>
  <remarks>Jomo Joshua Kiplimo</remarks>
  <agentID>2448</agentID>
  <check>Wr9UGeNFGkP7JgZzne5nlxK6x9I4mVOkxwzMF1gSqdUu2zVRICVvQfc-
tHnVfrOYkG2O5gisL/IPoS/ECXjuzWfu/qpHQzVFo4GG2YW/C6kuOC563BM5m-
DIlm3F2ULbHP/bJUHPrhNeCJGBKDC25h28bbklb50FBx5mavGS/GvaT2186hXJS-
F4gtvl8tN0WjOowW/6PIC+LSH+0cRGiKej0lpo03G2ptauTOkUBYeUbjvJLhPQkLiGFdis-
pZRGvsrTpfiHluQbxrjtIWY138M/Ozd2WOCyXSSVPzdD+yV5Hab0TdpT0+VkQD9AgS-
DSbTd0+GeMaeeKlpDAk3JmJRMjQ==</check>
  <fees>0.0</fees>
  <instCode>TZ</instCode>
  <tranAmount>20000.00</tranAmount>
  <messageID>21000q001</messageID>
  <currencyCode>TZS</currencyCode>
  <countryCode>TZ</countryCode>
  <bicCode></bicCode>
  <senderName>Andrii Ol Borysenko</senderName>
  <mobileNumber>255754000000</mobileNumber>
  <walletName>VODACOMTZ</walletName>
</ns2:tranApplyRequest>
</S:Body>
</S:Envelope>

```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>

```

```

<ns2:tranApplyRequestResponse
  xmlns:ns2="http://ws.equitybank/">
  <return>OK|167624122038</return>
</ns2:tranApplyRequestResponse>
</soap:Body>
</soap:Envelope>

```

3.2.5 Send Money to MTN UG

Request:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber></accountNumber>
      <paymentReference1>876q655</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
      <paymentReference3>876q655</paymentReference3>
      <paymentType>WorldRemit</paymentType>
      <remarks>Jomo Joshua Kiplimo</remarks>
      <agentID>2448</agentID>
      <check>GmYIYAyZCF0hNdaxHRE4Y7PIJ20LcsvdtJtM4A9IfEtLifg+JJizjFc-
VI655fkalmLHBhU3Eyf2nd17jYuxinaxgFQDCQ/sLhKtR65QI4hI2LN1LHQVWcE1I2OX-
QKSeqS7KcUrf3tF4ZkZc1dIPitIQhSOQTEgUOoEWHy/o44Q8xStMwAFn2qZWSDwApi-
XDPYkDm8cU8wMoPWhCq1UhRlxUD4V3qEPFsXW5EIPJe+I26evpQVgyFtvdeQwAjc/
OP2sBOaweSIElzlzLhUbgtxOIneAuMaWQDmz8mOTz9TEJ32ksBEcJs51/dxLkVmTQLR-
F8bBdRX+q9P1lwP4cu5mA==</check>
      <fees>0.0</fees>
    </ns2:tranApplyRequest>
  </S:Body>
</S:Envelope>

```

```

    <instCode>UG</instCode>
    <tranAmount>8014.55</tranAmount>
    <messageID>8767q5651</messageID>
    <currencyCode>UGX</currencyCode>
    <countryCode>UG</countryCode>
    <bicCode></bicCode>
    <senderName>Andrii Ol Borysenko</senderName>
    <mobileNumber>250722000000 </mobileNumber>
    <walletName>MTNUG</walletName>
  </ns2:tranApplyRequest>
</S:Body>
</S:Envelope>

```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|560927122223</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>

```

3.2.6 Send Money to MTN RW

Request:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
```



```

<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber></accountNumber>
      <paymentReference1>23a9982274</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
      <paymentReference3>23a9982274</paymentReference3>
      <paymentType>WorldRemit</paymentType>
      <remarks>Jomo Joshua Kiplimo</remarks>
      <agentID>2448</agentID>
      <check>Z4hg5OA4FycuUQq2queFTf2fvo0vahiylikL1VGrjCovRLpWd+jUo-
Uy2if52BdOnPLgAlxgswpUE+zEO9PJ/qKGOaaAEAPUY39mbl+QYlc05M7ec-
FK3PHNHMLoE6RhLmHeuwb9ztDWvQ1FYRcV+mAIRCg2Aagnz4WgmMNO8Hp-
G44GkAVCI66gzH83bCAqqtFFOwmZ6V1LM8dgWmEsLjRD4xZiwfNNA9gerPKXBs-
PEZ5k4ij5wwbHB89V8k1Kh8D+/250Q58aDwHq/NI+NjbEht0B1k9iPpB301G9BSi-
WdqhgY/dOynKXhm/Nr08A5YQZqO8WnHYRY5sIN+8+7EHyYw==</check>
      <fees>0.0</fees>
      <instCode>RW</instCode>
      <tranAmount>1000.55</tranAmount>
      <messageID>233455274</messageID>
      <currencyCode>RWF</currencyCode>
      <countryCode>RW</countryCode>
      <bicCode></bicCode>
      <senderName>Andrii Ol Borysenko</senderName>
      <mobileNumber>250788000000 </mobileNumber>
      <walletName>MTNRW</walletName>
    </ns2:tranApplyRequest>
  </S:Body>
</S:Envelope>

```

```
</S:Body>
</S:Envelope>
```

Response:

```
<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|981879122341</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>
```

3.2.7 Send Money to Tigo RW

Request:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<S:Envelope
  xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:ns2="http://ws.equitybank/">
  <S:Header xsi:type="xs:string"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  </S:Header>
  <S:Body>
    <ns2:tranApplyRequest>
      <accountNumber></accountNumber>
      <paymentReference1>8760s5tg6565</paymentReference1>
      <paymentReference2>Jomo Joshua Kiplimo</paymentReference2>
```

```

<paymentReference3>8760s5tg6565</paymentReference3>
<paymentType>WorldRemit</paymentType>
<remarks>Jomo Joshua Kiplimo</remarks>
<agentID>2448</agentID>
<check>Hg2/6iZ1vVkcwS0FKXvwSPzVL0ZSlcjsvo7XERQPgNK0yTYIPAgETp0Tc-
PQtBaOFY0LPNfJdTVjuYCVXeXJnjXMxkshx24legsbFwo6f0C8mb/v+YtFXDPihal-
AUP0e5EM+bWB7W8NnI7+KaOjP9EOh286XF+a7CRmEUlOZSZyNDIAbNOF2r5GgBU9l-
sQs0IvrjsU8RMfaNtjDY+7Cq9YmJ+waRBLLlGM/kZTn2Ral2wSG+eip0ABS4CZhFwT-
NCZYaCZiZl51ck1D9vsbOu27XUwrGxnAWNCvN9/orEK6b2Z7IM1cqXPq6aRd-
KOB6nuLj+RUJ9bGsH2PWkcP0PBgA==</check>
<fees>0.0</fees>
<instCode>RW</instCode>
<tranAmount>8014.55</tranAmount>
<messageID>876756yg07565</messageID>
<currencyCode>RWF</currencyCode>
<countryCode>RW</countryCode>
<bicCode></bicCode>
<senderName>Andrii Ol Borysenko</senderName>
<mobileNumber>250722000000 </mobileNumber>
<walletName>TIGORW</walletName>
</ns2:tranApplyRequest>
</S:Body>
</S:Envelope>

```

Response:

```

<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse
      xmlns:ns2="http://ws.equitybank/">
      <return>OK|536567122505</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>

```

```
</soap:Body>  
</soap:Envelope>
```

4. Monitoring

Both application logs and azure application insights are available.

For application logs, follow below steps:

4.1 Install both **azure-cli** and **kubernetes-cli**

4.2 Have with you the name of the deployments. In UAT, we have two deployments, **producer** and **imt-wr-consumer**

4.3 For each of the deployments, follow the below commands to get logs

a. *% kubectl get pods -l app=producer*

NAME	READY	STATUS	RESTARTS	AGE
producer-deployment-5b4ff5d67-vtrwf	1/1	Running	0	5d

b. *% kubectl logs -f producer-deployment-5b4ff5d67-vtrwf*

5. Certificate Generation

Steps:

1. Generating the key pair (Private and Public keys)

From your terminal:

```
~ % keytool -genkeypair -alias worldRemitSenderKeyPair -keyalg RSA -keysize 2048  
-dname "CN=WorldRemit_EquityBank" -validity 365 -storetype JKS -keystore /Users/  
bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitSender_keystore.jks  
-storepass WorldRemitJKSPass
```

Where:

1. `/Users/bishop/Programming/KeyStore/WorldRemit/ProdTest` is the path to your keystore
2. `worldRemitSender_keystore.jks` is the name of your keypair
3. `WorldRemitJKSPass` is the password
4. `worldRemitSenderKeyPair` is the keypair alias name.

OutPut:

Enter key password for <worldRemitSenderKeyPair>
(RETURN if same as keystore password):

Warning:

The JKS keystore uses a proprietary format. It is recommended to migrate to PKCS12 which is an industry standard format using "keytool -importkeystore -srckeystore /Users/bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitSender_keystore.jks -destkeystore /Users/bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitSender_keystore.jks -deststoretype pkcs12".

Note:

1. Ignore the warning
2. Hit Enter to use the same keystore password

2. Publishing the public Key

```
~ % keytool -exportcert -alias worldRemitSenderKeyPair -storetype JKS -keystore /  
Users/bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitSender_keystore.jks  
-file /Users/bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitSender_certificate.cer -rfc -storepass WorldRemitJKSPass
```

OutPut:

Certificate stored in file </Users/bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitSender_certificate.cer>

Warning:

The JKS keystore uses a proprietary format. It is recommended to migrate to PKCS12 which is an industry standard format using "keytool -importkeystore -srckeystore /Users/bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitSender_keystore.jks -destkeystore /Users/bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitSender_keystore.jks -deststoretype pkcs12"

Note:

1. Ignore the warning

3. Loading a Public Key

```
~ % keytool -importcert -alias worldRemitReceiverKeyPair -storetype JKS  
-keystore /Users/bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitReceiver_keystore.jks -file /Users/bishop/Programming/KeyStore/WorldRemit/ProdTest/worldRemitSender_certificate.cer -rfc -storepass WorldRemitJKSPass
```

OutPut:

```
Owner: CN=WorldRemit_EquityBank
Issuer: CN=WorldRemit_EquityBank
Serial number: 309d6711
Valid from: Thu Dec 10 14:11:21 EAT 2020 until: Fri Dec 10 14:11:21 EAT 2021
Certificate fingerprints:
    MD5: 30:69:E9:35:13:CE:98:5C:A5:C2:47:AE:06:74:E5:FF
    SHA1: E7:93:F6:29:F1:61:C4:04:96:DB:A4:86:62:FA:C8:F4:84:17:EB:F6
    SHA256:
10:82:B1:ED:D6:6E:F7:2F:37:2E:01:9C:1B:90:6E:3A:EC:73:DD:45:E7:0C:7B:68:C4:D5:69:3A
:09:40:CE:8C
Signature algorithm name: SHA256withRSA
Subject Public Key Algorithm: 2048-bit RSA key
Version: 3
```

Extensions:

```
#1: ObjectId: 2.5.29.14 Criticality=false
SubjectKeyIdentifier [
KeyIdentifier [
0000: 3B A5 E5 F2 6F 71 68 5B    8C 3D 12 03 0D 02 17 08    ;...oqh[.=.....
0010: C9 52 95 AA                .R..
]
]
```

```
Trust this certificate? [no]: yes
Certificate was added to keystore
```

Note:

1. Enter yes to trust the certificate

4. Expected Result

Share the generated jks, and store the certificate in GAG.

The keystore folder should look like this:

Favourites	Name	Date Modified	Date Created	Size	Kind	Date Last Open
Applications	worldRemitReceiver_keystore.jks	Today at 14:18	Today at 14:18	821 bytes	Bracke...ument	--
AirDrop	worldRemitSender_certificate.cer	Today at 14:12	Today at 14:12	1 KB	certificate	--
Recents	worldRemitSender_keystore.jks	Today at 14:11	Today at 14:11	2 KB	Bracke...ument	--
Desktop						
Downloads						
Documents						
Movies						
Music						
Pictures						
bishop						

Also use, SALT=810057ed7d654cda8c5cf523ed8890jmc